

# Abdullah Khaled

Robotics & Autonomy Engineer-in-Training

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## Education

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**Wakeland High School**, High School Diploma – Frisco, TX Aug 2023 – May 2027

- Unweighted GPA: 4.0
- Math readiness: AP Calculus AB & BC; planned Multivariable Calculus
- Science readiness: AP Physics 1; AP Biology; planned AP Physics 2 and AP Physics C
- 18 AP courses by graduation (completed and planned)

**Frisco ISD Career and Technical Education Center**, Project Lead The Way – Frisco, TX Aug 2023 – May 2027

- Nationally Distinguished PLTW Engineering Program
- Courses: PLTW Introduction to Engineering Design (9th grade), PLTW Principles of Engineering (10th grade), PLTW Digital Electronics (11th grade), PLTW Engineering Design and Development (12th grade)
- Engineering Portfolio: [akhaled247.github.io/pltw-engineering](https://akhaled247.github.io/pltw-engineering)

## Technical Experience

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**Chief Technical Advisor**, ITKAN Robotics (FRC 9128) – Frisco, TX Aug 2025 – present

- Provided high-level technical oversight across all robot subsystems for a state-level team of 40 members
- Built an extensive publish-subscribe telemetry and motor logging framework with match logs viewable after events for diagnosing issues and debugging autonomous routines
- Led development of robot state estimation and localization using sensor fusion of AprilTag vision and swerve-drive odometry, achieving <2 inch positional error
- Engineered autonomous control architecture with finite-state-machine (FSM) logic, Dijkstra-based path planning on time-weighted graphs, and PID-aligned speed/control tradeoffs
- Used WPILib Java mechanism simulation to validate autonomous and mechanism behaviors ahead of build to reduce integration risk
- Maintained and extended a >1000-line Java codebase spanning autonomous routines, teleoperated control, vision, and mechanisms using WPILib's command-based framework
- Achieved 100% autonomous success rate at the first event, produced the 5th-highest-scoring autonomous routine worldwide, and won an autonomous award
- Provided resources to 1000+ people online via [mantik.netlify.app](https://mantik.netlify.app) and an associated YouTube playlist

**Software Lead**, ITKAN Robotics (FRC 9128) – Frisco, TX Dec 2024 – July 2025

- Provided technical leadership on the organization of software team projects, educating and training new programming team members, and maintaining the team's FRC codebases
- Taught 5–10 students Java programming with a command-based subsystem architecture; trained debugging/troubleshooting and mechanism control using control theory
- Taught control theory concepts using PID and motion profiles, and used mechanism simulation to validate constraints before events
- Implemented autonomous Bezier pathing using a command-based architecture to coordinate motion profiles, mechanism actions, and sensor feedback
- Advanced the team to Texas State Championship playoffs (89 teams)
- Robot codebase and documentation: [akhaled247.github.io/first/reefscape](https://akhaled247.github.io/first/reefscape)

**Programming Mentor**, ITKAN Robotics of ICF (FTC 26357, 28931) – Frisco, TX

Sept 2024 – Feb 2026

- Implemented Kalman-filtered state estimation combining AprilTag vision and wheel odometry for real-time robot localization and automated turret targeting
- Mentored a full all-rookie FTC team through the engineering design cycle from prototyping and debugging to competition, emphasizing measurement-driven iteration
- Mentored refugee students in FLL workshops and led weekly summer training for FTC and FRC students using self-developed OARobotics curriculum ([akhaled247.github.io/personal/oarobotics](https://github.com/akhaled247/personal/oarobotics))

**Co-Founder, President**, Wakeland High School Wolverine Robotics (FTC 33791) – Frisco, TX

Aug 2024 – present

- Co-founded Wakeland robotics club, growing membership from 0 to 50+ students in the first year
- Secured funding and constructed competition-ready robot in under 2 weeks
- Teach CAD software (Onshape) and Java programming through monthly technical workshops
- Manage club operations including mission development, curriculum design, and budget management

**Math and Reading Instructor**, Gideon Math and Reading Center – Frisco, TX

June 2024 – present

- Instruct K-11th grade students in mathematics (Algebra, Geometry), reading, and grammar
- Analyze student performance data to identify learning gaps and develop personalized plans

**Math and Reading Teaching Assistant**, Gideon Math and Reading Center – Frisco, TX

Jan 2022 – May 2024

- Supported instruction and grading workflows, improving grading efficiency by approximately 50%

## Projects

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**OralVision – The Conrad Challenge**

Oct 2025 – present

- Developed a privacy-preserving oral disease screening pipeline with centrally trained model + on-device inference, without transmitting raw patient images (HIPAA-aligned federated learning framing)
- Designed, wired, and programmed the imaging device (XIAO-ESP32-C6 MCU + OV5642 5MP camera), including power-aware capture constraints (camera current draw can trip the 5V voltage regulator)
- Bench-tested image capture quality and robustness across lighting/viewing angles to maximize reliable low-cost imaging
- Verified repeatability by running multiple capture sessions under comparable conditions and confirming output consistency
- Iterated through 3 PCB revisions and 5 3D-printed snap-fit case revisions to improve reliability and manufacturability
- Used Onshape 3D CAD software to design and 3D print the device, reducing manufacturing costs
- Awarded Conrad Challenge Innovator Award and Alternate Finalist; Diamond Challenge Semi-Finalist
- Patent application in progress with the United States Patent and Trademark Office (USPTO)

**WatchFall – Samsung Solve for Tomorrow**

Sept 2025 – present

- Designed fall-detection device using Raspberry Pi Pico with on-device TensorFlow Lite inference
- Developed lightweight ML model for real-time motion tracking and fall detection with buzzer alert system
- Engineered a hybrid dataset combining public and custom annotated data, reducing false positive rate
- Integrated an Arducam HM01B0 Monochrome QVGA SPI Camera Module with Raspberry Pi Pico for resource-aware frame capture
- Planned next validation: evaluate false positives and alert responsiveness on recorded motion/fall scenarios, then tune fall rules and thresholds before expanded on-device testing

## Skills

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**Robotics, ML & Vision:** TensorFlow, TensorFlow Federated, OpenCV, SolvePnP, AprilTag vision, Swerve-drive odometry, Kalman Filtering, Finite State Machines, Dijkstra path planning, PID control, Mechanism Simulation, Control Theory, Federated Learning

**Software & Data:** Python, Java, JavaScript, R, Google Sheets, WPILib (command-based)

**CAD, Electronics & Prototyping:** Fusion 360, Onshape, NI Multisim, KiCAD, EasyEDA, Soldering

**Web & Productivity:** HTML, CSS, Jekyll, Google Workspace, Microsoft Office

## Leadership & Activities

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**Vice President**, Mu Alpha Theta/Math National Honor Society Aug 2024 – present

- Lead organization as Vice President, managing budget for 100+ members
- Expanded Math Lab Program and coordinated peer tutoring, attaining over 20% member participation
- Organized annual middle school math competition with 100+ participants, 3 participating schools, and 35+ volunteers to promote STEM

**Treasurer**, Mu Alpha Theta/Math National Honor Society Aug 2023 – July 2024

- Designed and implemented Math Lab Program coordinating student-led Algebra I tutoring with teachers

**Co-Founder, Secretary**, Muslim Student Association (MSA) – Wakeland High School, Frisco, TX Sept 2024 – present

- Developed mission statement, recruited teacher sponsor and members, and presented activities to school administration
- Negotiated with school administration to establish a dedicated prayer room, coordinating facility setup and policies

**Lead Math and English Instructor**, Ma'ruf Project Taleem – Dallas, TX Sept 2023 – present

- Teach Math and English to refugee students and mentor them in academics, time management, and social integration
- Coordinate small-group instruction and parent communication to support students' long-term academic progress

## Honors & Awards

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- (2026) OralVision: Conrad Challenge Innovator Award & Alternate Finalist; Diamond Challenge Semifinalist
- (2026) FTC 33791: Event Winner; Control Award Winner (FiT-North Dallas Semi-Regional, 25 teams; FiT-North U-League)
- (2025) FRC 9128: FIRST® Leadership Award Semifinalist; Event Winner (FiT Victoria, 25); Autonomous Award (FiT Plano, 31); Creativity Award (FiT Amarillo, 36)
- (2025) FTC 26357: Finalist Alliance Captain (FiT-North McKinney Qualifier, 19 teams)
- (2025) Collegeboard: AP Scholar with Distinction; National Recognition Program School Recognition Award